Functional Characterization and Reference Genome Assembly of *Acanthamoeba polyphaga* (ATCC-30461) – T4 genotype.

Karthikeyan Govindan [1], Blossom Benny [1], Venkata Raghava Mohan [2], Dilip Abraham [1]

1. Wellcome Trust Research Laboratory, Division of Gastrointestinal Sciences, Christian Medical College, Vellore, Tamil Nadu, 632004
2. Department of Community Health and Development, Christian Medical College, Vellore, Tamil Nadu, 632002

Email - karthikeyan.g@cmcvellore.ac.in

Background

Acanthamoeba are ubiquitous protozoa found in diverse soil and water environments. They can act as opportunistic pathogens and serve as protective niches for various bacterial pathogens outside the host, facilitating the survival and proliferation of these pathogens. Comprehensive reference genomes with detailed genomic and functional characterizations are limited, which poses challenges in studying host-pathogen interactions by using these larger protists as a model.

Methods

Enrichments

Acanthamoeba polyphaga ATCC-30461 was cultured in PYG medium to obtain a monolayer with a concentration of 3 × 10^6 cells/mL.

Whole Genome Sequencing

DNA extraction - QIAamp DNA Mini Kit

Library preparation - Nextera XT kit

Sequencing platform - Illumina MiSeq (2 × 300)

Metatranscriptomics

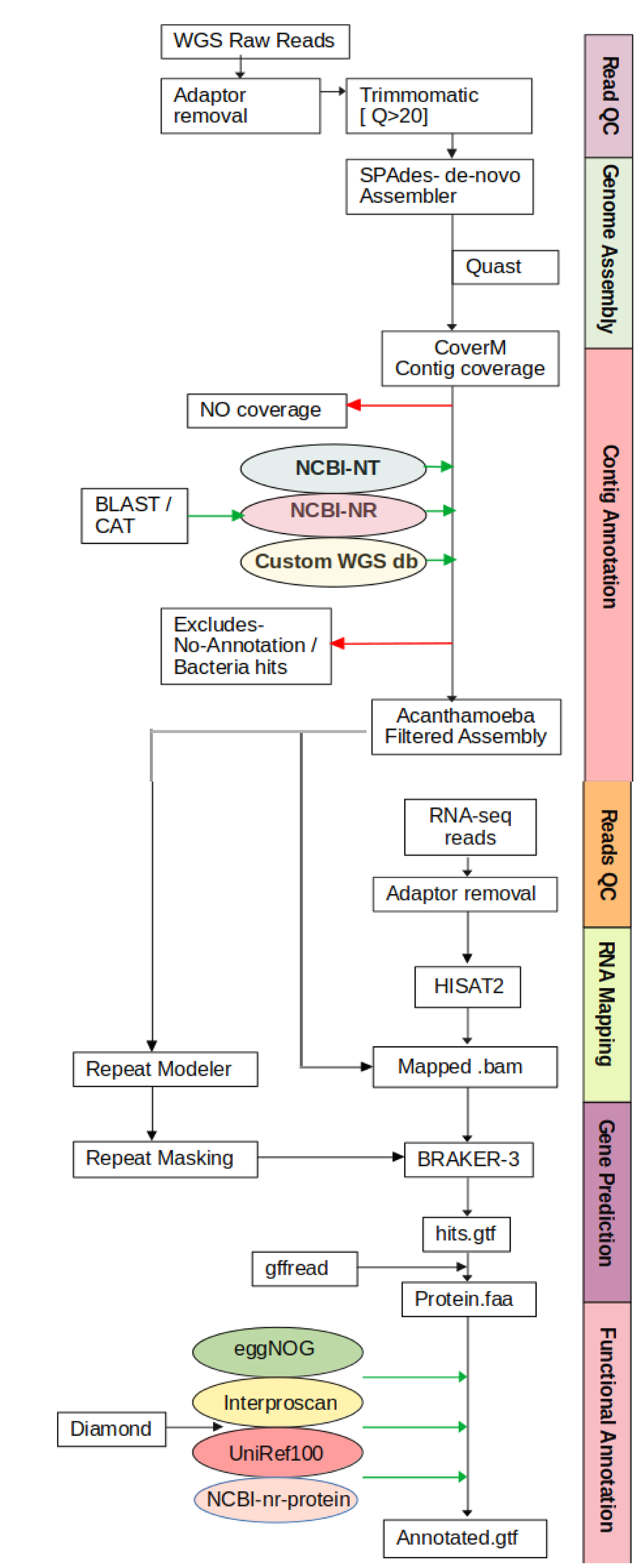
RNA extraction - QIAamp viral RNA minikit

Library preparation - Modified NEBNext RNA Ultra II kit

Sequencing platform - Illumina NovaSeq 6000 (2 ×150)

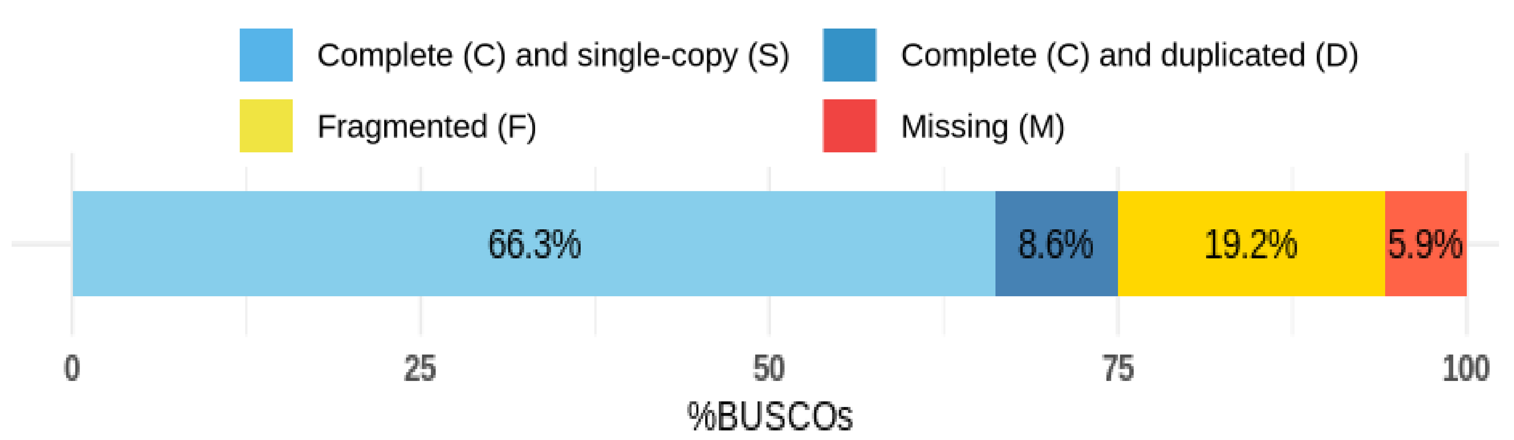
Results

Workflow for Constructing a Reference Genome

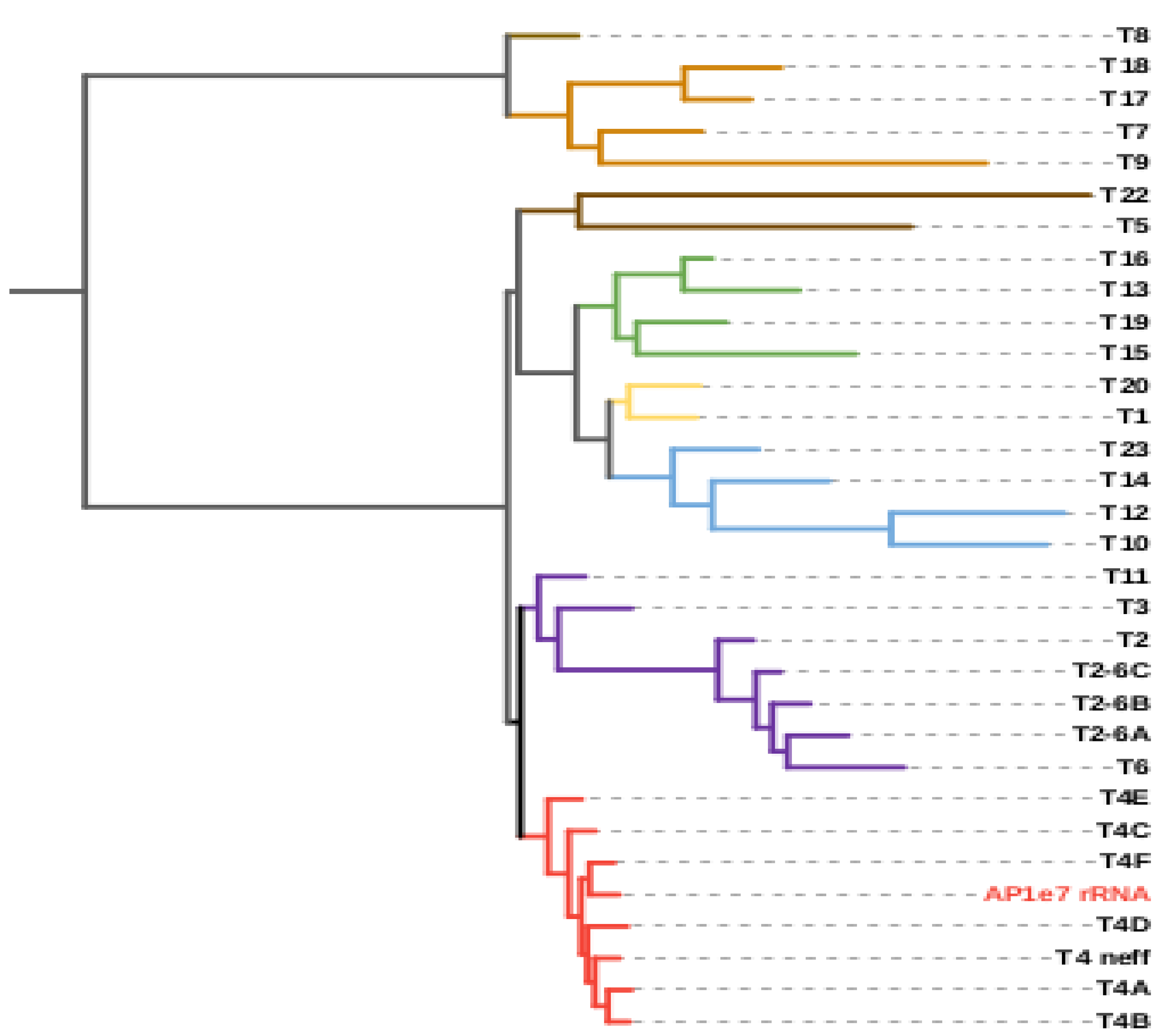


- Assembled contigs were taxonomically classified using BLAST (core-NT) and CAT, with additional annotation from a custom *Acanthamoeba* WGS database (28 genomes downloaded from NCBI)
- Contigs with *Acanthamoeba* hits in any of these database were included, while those with bacterial hits or no hits were excluded.
- Gene prediction in BRAKER3 using repeat-masked filtered contigs and mapped RNA reads resulted in 22,067 genes and 23,998 transcripts, including 1,931 alternative transcripts.
- The predicted proteins were assessed for gene completeness in BUSCO using busco lineage eukaryota_odb10

BUSCO Assessment Results

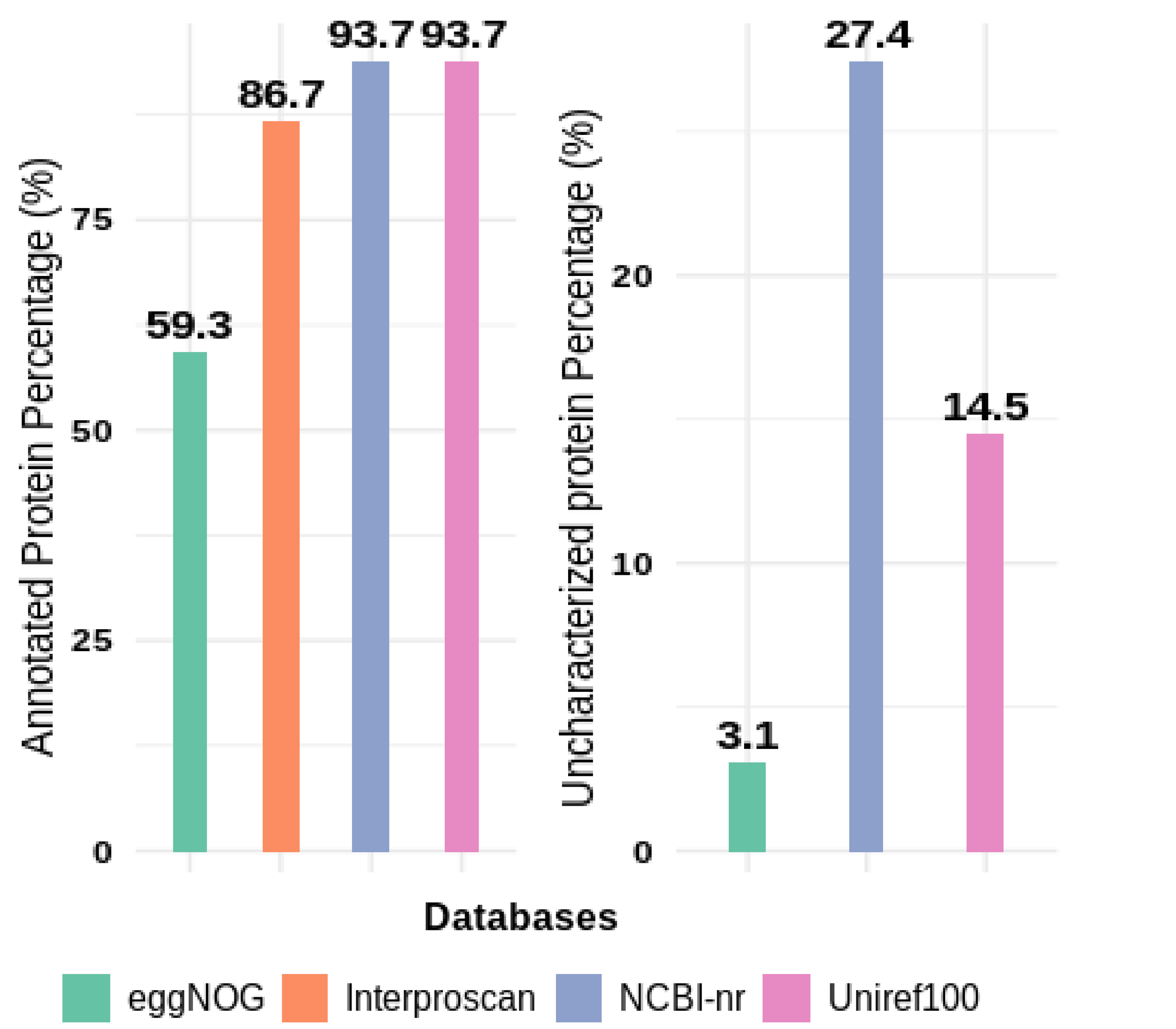


18s rRNA Phylogenetic tree



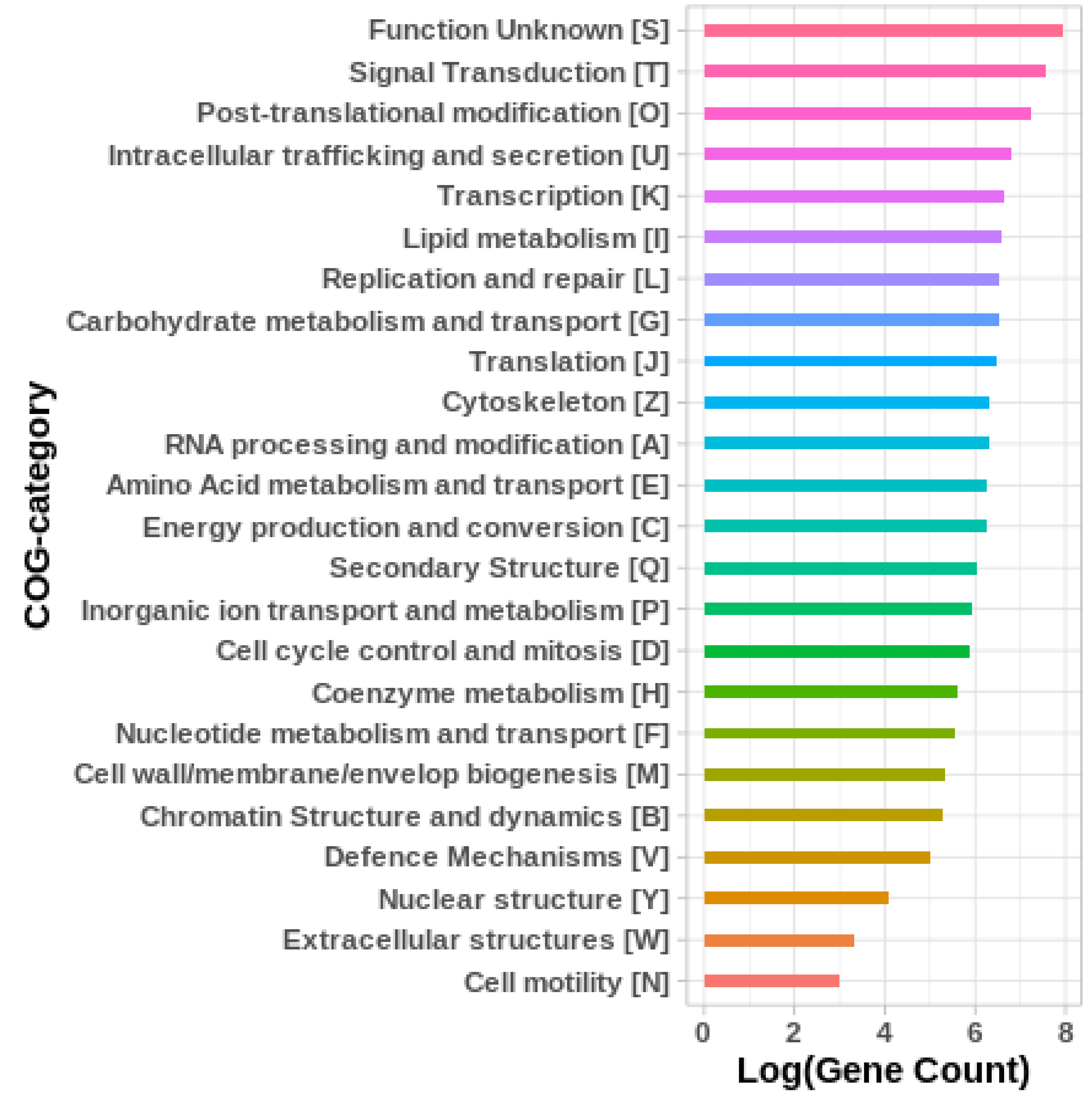
- The comparison of *A.polyphaga* with *A. castellanii_strNEFF* using OrthoFinder showed 1,126 unique orthologous groups, which included 2,891 genes in *A. polyphaga* ATCC-30461.
- The phylogenetic analysis with 23 different *Acanthamoeba* genotypes showed that ATCC-30461 (AP1e7 rRNA) falls within the T4 genotype and is closely related to the T4F subgenotype.

Comparison of Protein Functional Databases



- Comparisons between functional databases revealed that both UniRef100 and NCBI-nr had more annotations, with NCBI-nr containing a greater number of uncharacterized proteins compared to UniRef100.
- EggNOG-COG functional analysis indicated that genes related to cellular processing and metabolism were more prevalent

eggNOG Predicted Gene Functions



Conclusion

- The annotated proteins of ATCC-30461 could serve as a reference for annotating other *Acanthamoeba* species and studying gene functions in pathogen interactions.
- In addition to short-read sequencing, long-read WGS and transcriptomics are required for chromosome-level assembly and accurate transcript structure, as 20% of genes were fragmented (based on BUSCO results).
- Approximately 40% of genes lack orthologous information, while 14.5% remain uncharacterized in UniRef100. Structure-based homology prediction can be employed as an additional method for functional annotation of this unknown proteins.

Talk to us on Slack !

References